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The 13-year long-term follow-up on the effectiveness and immunogenicity of the quadrivalent human papillomavirus vaccine in Chinese females vaccinated at 20–45 years of age

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ABSTRACT

The quadrivalent human papillomavirus (qHPV) vaccine demonstrated high efficacy against vaccine-type-related precancers in Chinese females through a 6.6-year double-blind, randomized control trial (V501–041, NCT00834106). We present results for its long-term follow-up (LTFU) phase, evaluating vaccine effectiveness and immunogenicity throughout 13 years post Dose 1. Participants receiving qHPV vaccines at ages 20–45 in the V501–041 study constituted the early vaccination group (EVG). Those who received the placebo in V501–041 but were vaccinated during 2018–2020 formed the catch-up vaccination group (CVG), while the control group (CG) comprised those remaining unvaccinated throughout V501–041 and LTFU. The primary endpoint was histopathologically-confirmed cervical intraepithelial neoplasia (CIN) 2+ tested PCR-positive for HPV-16/18. Antibodies against HPV-6, 11, 16, and 18 were assessed using an immunoglobulin G (IgG) Luminex immunoassay at year 13. A total of 1100 participants from V501–041, of whom 978 were followed in LTFU, were included with a median follow-up of 152.2 months. No cases of the primary endpoint were observed in the EVG during 5513.6 person-years of follow-up, while one case was observed in the CG over 2951.4 person-years of follow-up, indicating a 100% risk reduction in the EVG compared to the CG. In the CVG, no cases were detected from vaccination to the latest follow-up (0.0 per 10 000 person-years), contrasting with an incidence of 22.3 per 10 000 person-years prior to vaccination. Seropositivity rates at year 13 remained >92% (for HPV-6/11/16) and 84% (HPV-18). These findings demonstrated continued vaccine protection against cervical precancers and sustained immunogenicity through 13 years post-vaccination.

Trial Registration: The LTFU study was registered at the Chinese Clinical Trial Registry, ChiCTR2100052313.

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Introduction

Cervical cancer is a global health issue with most fatalities occurring in low-income and middle-income countries (LMICs).[1] China, accounting for 17.3% of the global female population,[2] contributed to 22.8% of new cases and 16.0% of deaths worldwide in 2022.[3] Prophylactic human papillomavirus (HPV) vaccines have been proven safe and highly effective against cervical HPV infection and related precancers.[4] However, given the lifelong risk of HPV exposure,[5] HPV vaccines should provide continued protection to maximize vaccination benefits. Accordingly, the World Health Organization (WHO) recommends incorporating long-term follow-up (LTFU) studies to evaluate vaccine protection.[6]

Studies from several countries have supported the sustained protection and safety of HPV vaccines, with no breakthrough precancerous lesions related to vaccine types through 10–12 years of follow-up.[7–16] Yet, most of these studies focus on females vaccinated at 9–26 years of age,[7–12,14–16] The 10–12 years of follow-up may not be sufficient to observe vaccine breakthrough cases since it does not encompass the age range for the highest risk of cervical precancers or cancers.[17] In addition, while the current catch-up vaccination strategy primarily targets females aged ≤26 years,[18] extending this age limit is of growing concern, as females over 26 years also face ongoing risks of disease and infection throughout their lives.[5,19] Population-based vaccine evidence for this age group may help optimize vaccination decisions.

In China, local population-based data remain underreported regarding long-term vaccine protection. Furthermore, in the absence of nationwide HPV vaccination programs, vaccine coverage was below 5% and 15% among females aged 9–14 and 15–45 in 2022, respectively.[20] LTFU evidence is of

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urgent need and may be valuable for public confidence in vaccines and the implementation of expanded immunization, consequently contributing to global cervical cancer elimination.

Therefore, to assess the long-term effectiveness and immunogenicity of HPV vaccines in Chinese females, a 13-year LTFU study was conducted. Participants were recruited from an accomplished phase III randomized clinical trial (V501–041 [NCT00834106]), which had demonstrated the efficacy and safety of quadrivalent HPV (qHPV) vaccine in females vaccinated at ages 20–45 through 6.5 years of follow-up. The extension follow-up of V501–041 participants ensures a continuous and precise evaluation of vaccine breakthrough cases, as it covers the age range for the peak prevalence of cervical precancers (i.e., 40–49 years)[21] in Chinese females.

Patients and methods

Study design and participants

The V501–041 study was a multicenter, double-blind, randomized, placebo-controlled trial conducted at six sites in China. Eligible participants were sexually active females aged 20–45 years. Beginning in January 2009, participants were randomly assigned (1:1) to receive three doses of qHPV vaccine or placebo on Day 1, Month 2, and Month 6, with follow-up for efficacy assessment lasting 78 months and safety assessment lasting 90 months. The V501–041 study was completed in September 2016. Nearly half of the participants in the placebo group were offered catch-up vaccination with three doses of the qHPV vaccine from 2018 to 2020. Detailed study procedures and results have been published previously.[22,23]

In the present LTFU study, participants previously enrolled at one of the six study sites of the V501–041 study (i.e., Shanxi Province) were recalled. Participants initially receiving the qHPV vaccine in the V501–041 study were regarded as the early vaccination group (EVG). Those who initially received the placebo but were catch-up vaccinated from 2018 to 2020 constituted the catch-up vaccination group (CVG), while the control group (CG) comprised those remaining unvaccinated during both the V501–041 and LTFU phases. Two extended follow-up visits were conducted at approximately 10- (October 2019) and 13 years (January 2022) post Dose 1. At each follow-up, general informed consent, blood sampling (for EVG only), and cytology-based cervical screening were performed.

Study procedures

During LTFU, gynecological examination and cervical sampling (ThinPrep) were performed using the same methods as in the V501–041 study, as previously reported.[22,23] Briefly, Bethesda cytology results were reported by a central laboratory. Abnormal cytology included atypical squamous cells of undetermined significance (ASC-US) with concurrently high-risk HPV (HR-HPV) positive (detected by the Digene Hybrid Capture II^{TM}) or low-grade squamous intraepithelial lesion or higher-grade lesions (LSIL+). In case of abnormal cytology findings, participants were referred to colposcopy. According to the colposcopy findings, cervical biopsy or/and endocervical curettage (ECC) were taken if necessary. The biopsy or ECC specimen was formalin-fixed and paraffin-embedded for biopsy diagnosis by physicians from the central laboratory. The presence of vaccine and non-vaccine HPV types on the histopathologically-confirmed CIN 1+ specimen was detected using the LiPA HPV Genotyping PCR Assay, which can detect 14 hR-HPV types (HPV types 16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 66 and 68) and 11 LR-HPV types (HPV types 6, 11, 34, 40, 42, 43, 44, 53, 54, 70, 74).

On Day 1 of the V501–041 study, serum was collected in all participants for population identification using a competitive Luminex immunoassay (cLIA). At the 13-year visit, serum was collected from EVG participants for immunogenicity analyses using IgG Luminex immunoassay (IgG LIA)[24,25] (Figure 1).

Endpoints and analysis sets

Primary endpoints included the effectiveness of the qHPV vaccine on histopathologically confirmed CIN 2+ (included CIN2/3, adenocarcinoma in situ (AIS), or cervical cancer) tested PCR-positive for HPV-16/18. In addition, primary endpoints included immune persistence, summarized by IgG LIA seropositivity rates and geometric mean titers (GMTs) of anti-HPV-6, 11, 16, and 18 antibodies, respectively. Seropositivity was defined as an antibody titer greater than or equal to the assay lower limits of quantitation: 2 milli-Merck units per milliliter (mMU/mL) (for HPV types 6 and 11), 4 mmU/mL (HPV-16), and 3 mmU/mL (HPV-18).

Secondary endpoints included the vaccine effectiveness on histopathologically confirmed CIN 2+ tested PCR-positive for HPV-6/11/16/18, as well as effectiveness on histopathologically confirmed CIN 1+ tested PCR-positive for HPV-16/18 or HPV-6/11/16/18.

The per-protocol effectiveness (PP-E) population was the primary effectiveness analysis population, including participants who 1) received three doses of qHPV vaccine or placebo within 1 year, 2) were seronegative by cLIA on Day 1 and remained PCR-negative from Day 1 through Month 7 for the HPV type being analyzed, and 3) did not violate the protocol in ways that could affect the evaluation of effectiveness (definition of protocol violation is presented in Supplementary Table 2.1–2.2). Effectiveness was also analyzed in the modified Intention-to-Treat (mITT) population, comprising all participants who 1) received at least one vaccine or placebo dose, 2) were seronegative by cLIA and PCR-negative for the HPV types being analyzed on Day 1 of the V501–041 study (for CVG during the V501–041 period) or seronegative for HPV types being analyzed on Day 1 of the V501–041 study and remained PCR-negative for the considered HPV types from Day 1 to the last visit of the V501–041 study before receiving the catch-up vaccination (for CVG during the LTFU period), and 3) had at least one long-term follow-up visit. A detailed definition of the mITT population by vaccination group is described in the supplementary material.

The per-protocol immunogenicity (PPI) population was the primary immune persistence analysis population, including participants who 1) received three doses of qHPV vaccine within acceptable day ranges (administered on Day 1,

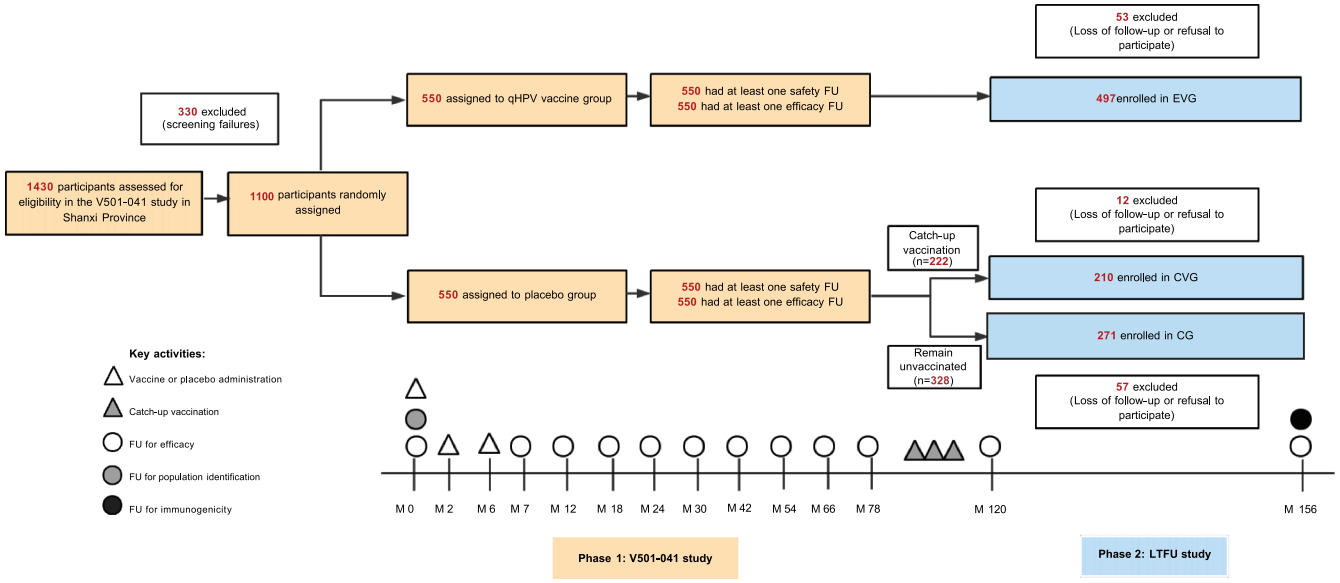


Figure 1. Study design and participant flow. CVG = catch-up vaccination group; CG = control group; EVG = early vaccination group; HPV = human papillomavirus; LTFU = long-term follow-up; qHPV = quadrivalent HPV

Month 2 [±3 weeks], and Month 6 [±4 weeks], respectively) during the V501–041 study, 2) were seronegative on Day 1 and remained PCR-negative from Day 1 through Month 7 for the HPV type being analyzed, and 3) did not violate the protocol in ways that could affect the evaluation of immunogenicity (definition of protocol violation can be found in Supplementary Table 2.5). Immune persistence was also evaluated in the all type-specific naïve participants with serology (ANPS) population, comprising participants who 1) received three doses of qHPV vaccine during the V501–041 study and 2) were seronegative on Day 1, and remained PCR-negative from Day 1 through Month 7 for the HPV type being analyzed.

Statistical analyses

Demographic and baseline characteristics of participants in the EVG, CVG, and CG were summarized, including the mean (SD) and median (range) age, age at first sexual intercourse, number of sexual partners, abnormalities on cervical cytology, and seropositive or PCR-positive to any of the four vaccine-targeted HPV types.

Effectiveness endpoints were presented separately for the EVG and CG during the V501–041 and LTFU period in the PP-E population and for CVG in the mITT population. The criteria for the per-protocol analysis in CVG could not be met, as no PCR or serology testing was conducted to detect HPV infections that occurred between the final V501–041 visit and the initiation of catch-up vaccination. The incidence of effectiveness endpoints per 10 000 person-years of follow-up and the exact Poisson 95% CIs were reported. Vaccine effectiveness was calculated as 100× (1-relative risk), where relative risk was the ratio of the EVG incidence relative to the CG incidence. The 95% CI of vaccine effectiveness was calculated using the exact method.[26] Effectiveness was also analyzed by age subgroups, including participants aged 20–26 and 27–45 years at the baseline of the V501–041 study.

Immune persistence was evaluated for the EVG in the PP-I and ANPS populations at year 13. IgG antibody seropositivity rates and GMTs with 95% CIs were presented for HPV-6, 11, 16, and 18. The 95% CIs for seropositivity rates were calculated based on the Clopper-Pearson exact method. GMTs were exponentiated mean natural log-transformed titers, and the corresponding 95% CIs were calculated based on the *t*-distribution.

All data analyses were performed using SAS version 9.4 (SAS Institute).

Results

Participant characteristics

The V501–041 study assessed 1430 participants for eligibility from the Shanxi Province, and 1100 of them were randomly assigned to the qHPV vaccine group and placebo group at a 1:1 ratio. During the vaccination phase, 550 participants received at least one dose of qHPV vaccine, while 550 received at least one dose of placebo. Of the 1100 participants in the V501–041 study, 978 agreed to participate in the LTFU study, with 497, 210, and 271 individuals included in the EVG, CVG, and CG, respectively (Figure 1). The median follow-up time was 152.2 months (interquartile range [IQR]: 150.8–154.9), which was calculated as the time interval from Dose 1 to the last follow-up of the LTFU study (Table 1)

The demographic and baseline characteristics were generally comparable among EVG, CVG, and CG at enrollment in the V501–041 study. The median age of all participants was 25 years (range 20–45 years). At the first LTFU visit (year 10), age and sexual behaviors were also comparable among EVG, CVG, and CG participants (Supplementary Table S1). Participants

Table 1. Baseline characteristics of participants in Shanxi Province at the time of Dose 1 in the V501-041 study and follow-up characteristics of the entire study.

	Early vaccination group (N = 551)		Catch-up vaccination group (N = 222)		Control group (N = 327)		Total (N = 1100)	
	n	(%)	n	(%)	n	(%)	n	(%)
Age (Years)								
20 to 26	397	72.0	163	73.4	231	70.6	791	71.9
27 to 45	154	28.0	59	26.6	96	29.4	309	28.1
Mean	27.4		26.7		28.3		27.6	
SD	5.7		4.2		6.9		5.8	
Median (min-max)	25 (20–45)		25 (21–36)		25 (20–45)		25 (20–45)	
Age at first sexual intercourse (Years)								
Mean	20.6		20.6		20.6		20.6	
SD	1.9		1.8		2.0		1.9	
Median (min-max)	21 (14–28)		21 (16–25)		21 (15–27)		21 (14–28)	
Number of sexual partners								
1	463	84.0	197	88.7	276	84.4	936	85.1
≥2	88	16.0	25	11.3	51	15.6	164	14.9
Abnormality in cervical cytology								
Yes	56	10.2	14	6.3	25	7.6	95	8.6
No	471	85.5	196	88.3	285	87.2	952	86.5
Unsatisfactory	24	4.4	12	5.4	17	5.2	53	4.8
Positive to HPV6, 11, 16, or 18								
By serology*	138	25.0	56	25.2	79	24.2	273	24.8
By PCR*	42	7.6	30	13.5	28	8.6	90	8.2
By serology or PCR*	154	27.9	66	29.7	95	29.1	315	28.6
Total follow-up time per woman (Months)								
Median (IQR)	152.2 (150.9–154.9)		154.6 (151.2–154.9)		151.8 (128.4–154.8)		152.2 (150.8–154.9)	

*Combined prevalence on HPV6, 11, 16, or 18.

accounting for each analysis population by vaccination group are presented in Table 2 and Supplementary Table S2.1–S2.5.

Effectiveness

During the V501–041 and LTFU study, in the per-protocol analysis, no cases of CIN2+ related to HPV-16/18 during 5513.6 person-years of follow-up were observed in the EVG, while one case was observed in the CG over 2951.4 person-years of follow-up representing an incidence of 3.4 (95% CI: 0.1–18.9) per 10 000 person-years, consequently indicating a 100% (≤ −999, 100) risk reduction in the EVG compared to CG. In addition, no cases of HPV-6/11/16/18-related CIN 2+ or CIN 1+ were observed in the EVG, while one and three cases of HPV-6/11/16/18-related CIN 2+ and CIN 1+ were observed, respectively, in the CG representing the incidence of 3.4 (0.1–18.8) and 10.2 (2.1–29.7) per 10 000 person-years (Table 3).

The per-protocol effectiveness results are presented by age stratum in Supplementary Table 3.1–3.2. For young females vaccinated at ages 20–26 years, the HPV vaccine indicated 100% (95% CI −171.3, 100) risk reduction of HPV-16/18-related CIN 1+ and 100% (−170.1, 100) for HPV-6/11/16/18-related CIN 1+. For females vaccinated at ages 27–45 years, the vaccine indicated 100% (≤ −999, 100) risk reduction of HPV-16/18 or HPV-6/11/16/18-related CIN 2+, as well as 100% (≤ −999, 100) of HPV-16/18 or HPV-6/11/16/18-related CIN 1 + .

Regarding the mITT effectiveness analysis in CVG, no cases of HPV-16/18-related CIN2+ were detected after catch-up vaccination, while the incidence before that was 22.3 per 10 000 person-years (95% CI 4.6–65.1) in this group. In addition, no cases of HPV-6/11/16/18-related CIN 1+ or CIN 2+ were detected in CVG from catch-up vaccination to year 13, while the incidence before catch-up vaccination was 22.2 (4.6–64.9) per 10 000 person-years for CIN 2+ and 22.2 (4.6–64.9) for CIN 1+ (Table 4). Similar trends for the decreased incidence of CIN during the LTFU period were also found in the two age subgroups (females vaccinated at 20–26 or 27–45 years of age, Supplementary Table 4.1–4.2)

Immunogenicity

Immunogenicity was assessed in EVG participants at the year 13. For the PP-I population, the seropositivity rates of anti-HPV6, 11, 16, and 18 antibodies remained at 92.3%, 92.9%, 99.4%, and 83.6%, respectively. The GMTs of anti-HPV6, 11, 16, and 18 antibodies were 32.95, 23.46, 117.85, and 17.31 mmU/mL, respectively (Supplementary Table 5.1).

For the ANPS population, the seropositivity rates of anti-HPV6, 11, 16, and 18 antibodies remained at 92.4%, 93%, 99.2%, and 83.2%, respectively. The GMTs of anti-HPV6, 11, 16, and 18 antibodies were 33.11, 23.53, 117.82, and 17.14 mmU/mL, respectively (Supplementary Table 5.2).

Discussion

Through the 13-year LTFU, the EVG observed no breakthrough cases of CIN 2/3, AIS, or cervical cancer related to HPV-16/18 (primary endpoints), nor any grade of CIN, AIS, and cervical cancer related to HPV-6/11/16/18 (secondary

Table 2. Participants accounting for the effectiveness analysis population by vaccination group (PP-E population, V501–041 study + LTFU study).

	EVG (N = 551)	CG (N = 327)
Number of participants who received at least 1 injection[a]	551	327
Eligible for the PP-E Analysis Related to:		
HPV 6/11/16/18	493	273
HPV 16/18	490	272
Ineligible for the PP-E Analysis Related to:		
HPV 6/11/16/18	58	54
HPV 16/18	61	55
Reason for Ineligibility[b]		
General protocol violation	2	4
A pigmented nevus at vulvae existed prior to enrollment	1	0
Received incorrect clinical material or dose amount	1	0
Received non-study vaccination[c]	0	2
Received immunosuppressives, IgG, or blood products	0	2
Missed 2nd and 3rd vaccination	2	0
Missed 3rd vaccination	4	10
Without Day 1 swab PCR results within an acceptable day range[d]	6	7
Without Month 7 swab PCR results within an acceptable day range[d,e]	32	28
HPV 6 or 11 Positive by Serology or PCR[f]		
Serology and/or PCR positive on Day 1	12	15
PCR positive at or before Month 7	17	19
HPV 16 Positive by Serology or PCR[f]		
Serology and/or PCR positive on Day 1	19	8
PCR positive at or before Month 7	24	11
HPV 18 Positive by Serology or PCR[f]		
Serology and/or PCR positive on Day 1	7	2
PCR positive at or before Month 7	16	6

[a]Participants who did not receive at least 1 injection were excluded from all analysis populations. [b]Participants are counted once in each applicable exclusion category. A participant may appear in more than one category. [c]Includes live virus vaccines received within 21 days before or 14 days after study vaccination or inactivated or recombinant vaccines received within 14 days of study vaccination. [d]Includes participants who are missing at least one required swab sample or PCR result. [e]Among participants who received all 3 vaccinations. [f]Day 1 includes seropositivity or PCR positivity. Post-Day 1 includes PCR positivity only. Applies only to the analysis populations for the respective HPV type(s).

PP-E = per-protocol effectiveness; N = number of participants randomized to the respective vaccination group.

Table 3. Reduction in the incidence of histology abnormalities between EVG and CG (PP–E population).

	EVG (N = 551)			CG (N = 327)			
	Participants (n/N)	Person-years of follow-up	Incidence per 10 000 person-years (95% CI)	Participants	Person-years of follow-up	Incidence per 10 000 person-years (95% CI)	EVG vs CG risk reduction estimate (95% CI)*
HPV 16- and 18-related CIN2+							
V501–041 study	0/485	2715.0	0 (0.0–13.6)	1/271	1516.4	6.6 (0.2–36.7)	100 (≤ −999, 100)
LTFU study	0/447	2311.7	0 (0.0–16.0)	0/231	1184.7	0 (0.0–31.1)	NA
V501–041 study + LTFU study	0/490	5513.6	0 (0.0–6.7)	1/272	2951.4	3.4 (0.1–18.9)	100 (≤ −999, 100)
HPV 16- and 18-related CIN1+							
V501-041 study	0/485	2715.0	0 (0.0–13.6)	3/271	1514.9	19.8 (4.1–57.9)	100 (−35.0, 100)
LTFU study	0/447	2311.7	0 (0.0–16.0)	0/231	1184.7	0 (0.0–31.1)	NA
V501-041 study + LTFU study	0/490	5513.6	0 (0.0-6.7)	3/272	2940.2	10.2 (2.1–29.8)	100 (−29.0, 100)
HPV 6-, 11-, 16-, and 18-related CIN2+							
V501-041 study	0/488	2732.4	0 (0.0–13.5)	1/272	1522.2	6.6 (0.2–36.6)	100 (≤ −999, 100)
LTFU study	0/450	2327.7	0 (0.0–15.8)	0/232	1187.8	0 (0.0–31.1)	NA
V501-041 study + LTFU study	0/493	5549.6	0 (0.0–6.6)	1/273	2961.1	3.4 (0.1–18.8)	100 (≤ −999, 100)
HPV 6-, 11-, 16-, and 18-related CIN1+							
V501-041 study	0/488	2732.4	0 (0.0–13.5)	3/272	1520.7	19.7 (4.1–57.7)	100 (−34.7, 100)
LTFU study	0/450	2327.7	0 (0.0–15.8)	0/232	1187.8	0 (0.0–31.1)	NA
V501-041 study + LTFU study	0/493	5549.6	0 (0.0–6.6)	3/273	2950.0	10.2 (2.1–29.7)	100 (−28.6, 100)

n/N: n = number of endpoint cases; N = number of participants in the analysis population with follow-up in the indicated study phase.

*Refers to the percentage reduction in incidence in the EVG versus the CG during the indicated phase.

EVG = early vaccination group; CG = control group; PP-E = per-protocol effectiveness; HPV = human papillomavirus; CIN 1+ = cervical intraepithelial neoplasia 1 or worse; CIN 2+ = cervical intraepithelial neoplasia 2 or worse; LTFU = long-term follow-up.

Table 4. Reduction in the incidence of histology abnormalities in CVG (mITT population).

	CVG (N = 222)		
	Participants (n/N)	Person-years of follow-up	Incidence per 10 000 person-years (95% CI)
HPV 16- and 18- related CIN2+			
V501-041 study	3/217	1347.0	22.3 (4.6-65.1)
LTFU study	0/167	567.6	0 (0.0-65.0)
HPV 16- and 18- related CIN1+			
V501-041 study	3/217	1346.8	22.3 (4.6-65.1)
LTFU study	0/167	567.6	0 (0.0-65.0)
HPV 6-, 11-, 16-, and 18-related CIN2+			
V501-041 study	3/218	1351.2	22.2 (4.6-64.9)
LTFU study	0/155	528.8	0 (0.0-69.8)
HPV 6-, 11-, 16-, and 18-related CIN1+			
V501-041 study	3/218	1351.0	22.2 (4.6-64.9)
LTFU study	0/155	528.8	0 (0.0-69.8)

n/N: n = number of endpoint cases; N = number of participants in the analysis population with follow-up in the indicated study phase.

CVG = catch-up vaccination group; mITT = modified intention to treat; HPV = human papillomavirus; CIN 1+ = cervical intraepithelial neoplasia 1 or worse; CIN 2+ = cervical intraepithelial neoplasia 2 or worse; LTFU=long-term follow-up.

endpoints). Similarly, no primary or secondary endpoint cases were detected in the CVG from the time of catch-up vaccination to the last follow-up. Durable vaccine-induced IgG antibody responses were detected, with seropositivity rates exceeding 90% for HPV types 6, 11, and 16 and 80% for HPV-18. These results represent the observed persistence of both effectiveness and immunogenicity throughout 13 years.

The V501–041 study demonstrated that the qHPV vaccine was 100% efficacious against CIN 2+ caused by HPV-6/11/16/18 up to 6.5 years post-vaccination.[23] Previous data from an 8-year follow-up of a small cohort of V501–041 participants (n < 400) reported no breakthrough cases of CIN 2+ caused by the four vaccine HPV types.[27] The present LTFU study further strengthens this evidence by presenting no vaccine breakthrough CIN 2+ cases through a maximum follow-up of 13.4 years in 1100 participants, indicating observed 100% clinical effectiveness.

Previous studies of bivalent, quadrivalent, and nonvalent HPV vaccines from several countries have demonstrated continued protection against type-related CIN 2+ through 10–12 years of follow-up in females vaccinated at 9–45 years of age.[7–16–28] The longest available data, with a median follow-up of 11.9 years, are from Nordic countries, which reported 100% protection of the qHPV vaccine against CIN 2+ related to HPV-16/18 in females vaccinated at ages 16–23.[10] Despite these data, local evidence in China is still imperative since geographical differences in HPV genotype distribution might cause variations in vaccine protection across countries.[29]

From the time of catch-up vaccination to the 13-year visit (approximately four-year interval), no breakthrough precancers related to the vaccine types were detected in the CVG participants, despite their prior exposure to HPV. This finding aligns with a recent meta-analysis indicating that in HPV-seropositive females, vaccine efficacy was 87% against cervical lesions related to vaccine types.[30] Furthermore, post-excisional HPV vaccination has been demonstrated to be effective against CIN2+ recurrence.[31,32] Taken together, these findings support that vaccinated adults still experience long-term benefits from reduced disease incidence, regardless of their HPV exposure history, although pre-sexual vaccination is optimal.

In addition to the effect of HPV vaccination, cervical screening every 6 months during the V501–041 phase may have contributed to the absence of breakthrough precancers in both the CVG and EVG by identifying early-stage abnormalities. On this point, we compared the precancer incidence in these two groups with the CG, which underwent harmonized screening procedures but remained unvaccinated throughout the follow-up period. The observed incidence of precancer cases in the CG (3.4 [0.1–18.8] per 10 000 person-years) indicated that no breakthrough cases in the CVG and EVG could be attributed to vaccine protection. One thing to note is that regular follow-up cannot be ignored for HPV-infected individuals after HPV vaccination, as persistent infection is strongly associated with the onset and recurrence of cervical cancer.[33]

The qHPV vaccine elicited robust and prolonged antibody responses to the vaccine-targeted HPV types through 13 years post-vaccination. Over 90% of the EVG participants remained IgG seropositive for HPV-6, 11, and 16, while over 80% remained seropositive for HPV-18. Although the threshold of antibody titers required for clinical protection is uncertain, animal studies have demonstrated that even titers 100-fold lower than the threshold of standard antibody assays may confer protection.[34] Furthermore, the three-dose HPV vaccination has been demonstrated to induce HPV-specific memory B cells, enabling a rapid and robust antibody response upon re-exposure to HPV antigens.[35] The absence of HPV-18-related breakthrough cases in our study supports continued vaccine protection via lower-than-detectible IgG-LIA antibody titers or immune memory.

Our study has several strengths. First, to our knowledge, it represents the longest follow-up on the effectiveness of HPV vaccines, with a median follow-up of 12.7 years (the previous longest data reported from Nordic countries had a median follow-up of 11.9 years).[10] Second, in contrast to previous LTFU studies,[7–12,14–16] our study covered the age range for the peak risk of cervical precancers. Therefore, no breakthrough precancerous cases observed in our study strongly support the hypothesis that vaccines suppress the peak risk of precancers. Third, our study rigorously evaluated the same endpoints as those in the V501–041 trial and continuously assessed the same population, ensuring the reliability of our findings.

The main limitation of our study is the replacement of the original unvaccinated group (placebo group in the V501–041 period) with a new unvaccinated group (CG in the LTFU period) due to catch-up vaccination. As a result, the randomization between vaccinated and unvaccinated groups was not maintained during the LTFU phase. Yet, our effectiveness estimates remain justified, as the risks of HPV exposure (e.g., age and sexual behavior) were comparable between EVG and CG at the initiation of the LTFU.

Reporting the longest follow-up on HPV vaccine effectiveness, our study provides strong evidence for continued protection against cervical precancers related to HPV-6/11/16/18 and sustained immunogenicity through 13 years post-vaccination, indicating no need for a booster dose over this period. The findings could be instrumental in implementing nationwide HPV vaccination in China and other LMICs lacking such programs. Meanwhile, our findings also support the benefit of catch-up vaccination in individuals up to age 45, reaffirming the HPV-FASTER recommendation to extend the age limit of vaccination programs alongside screening to achieve faster cervical cancer elimination.[36] Future studies on vaccine benefits in individuals aged over 45 years might provide promising insights into reducing the cervical cancer burden, particularly in the context of the increasing global life expectancy.

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Disclosure statement

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All authors qualify for authorship in adherence with the ICMJE guidelines. The work reported in the paper has been performed by the authors, unless specified in the text. The contribution of each author is as follows: T.M.W: conceptualization; data curation; investigation; methodology; formal analysis; writing-original draft; writing-review & editing. X.Q.X: conceptualization; data curation; methodology; writing – review and editing. C.H.P: formal analysis; writing – review and editing. S.Y.H: conceptualization; data curation; methodology; supervision; writing – review and editing. H.Z: data curation; writing – review & editing. X.Z: data curation; writing – review & editing. Y.L.Q: conceptualization; funding acquisition; methodology; supervision; writing – review and editing. F.H.Z: conceptualization; data curation; funding acquisition; methodology; project administration; supervision; writing – review and editing.

Data availability statement

The data supporting this study's findings are available from the corresponding author upon reasonable request. The data are not publicly available because of privacy or ethical restrictions.

Ethics statement

The analysis of the V501–041 phase was based on the secondary use of de-identified data and was considered exempt by the ethical committee of Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College (CHCAMS/PUMC).

The study protocol of the LTFU phase was approved by the ethical committee of Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College (CHCAMS/PUMC). Written informed consent was obtained from all participants. The study was registered at the Chinese Clinical Trial Registry, number ChiCTR2100052313.

Main text introduction

Long-term follow-up studies on HPV vaccine protection are recommended by the WHO due to the lifelong risk of HPV exposure and associated cervical cancers. However, the relevant data are currently lacking in China. Our study assessed the 13-year effectiveness and immunogenicity of quadrivalent HPV (qHPV) vaccines in Chinese females. Analyses showed that the vaccination cohort observed no breakthrough cervical precancers, with seropositivity rates remaining >92%. Continued clinical protection and sustained immune responses were demonstrated for at least 13 years post-vaccination.

Paper presentation

The abstract of our effectiveness results was partially reported at the IPVC 2023 Congress orally: Tianmeng Wen, Xiaoqian Xu, Chenghao Pan, Shangying Hu, Fanghui Zhao. Effectiveness of the quadrivalent human papillomavirus vaccine in Chinese females: a 13-year long-term follow-up of a phase 3, double-blind, randomized control trial.

Abbreviations

ANPS	all type-specific naïve participants with serology
ASC-US	abnormal cytology included atypical squamous cells of undetermined significance
CVG	catch-up vaccination group
CIN	cervical intraepithelial neoplasia
CIs	confidence intervals

cLIA	competitive Luminex immunoassay
CG	control group
EVG	early vaccination group
ECC	endocervical curettage
GMTs	geometric mean titers
HR-HPV	high-risk human papillomavirus
IgG LIA	IgG Luminex immunoassay
IQR	interquartile range
LTFU	long-term follow-up
LSIL+	low-grade squamous intraepithelial lesion or higher-grade lesions
LMICs	low-income and middle-income countries
mITT	modified intention-to-treat
mMU/mL	milli-Merck units per milliliter
PP-E	per-protocol effectiveness
PP-I	per-protocol immunogenicity
qHPV	quadrivalent human papillomavirus
WHO	the World Health Organization

References

1. Singh D, Vignat J, Lorenzoni V, Eslahi M, Ginsburg O, Lauby-Secretan B, Arbyn M, Basu P, Bray F, Vaccarella S, et al. Global estimates of incidence and mortality of cervical cancer in 2020: a baseline analysis of the WHO global cervical cancer elimination initiative. Lancet Glob Health. 2023;11(2):197–206. doi:10.1016/S2214-109X(22)00501-0.
2. World Bank Group. Population, female. https://data.worldbank.org/indicator/SP.POP.TOTL.FE.IN.
3. Chen J, Zhang Z, Pan W, Song Y, Zheng L, Li L, Ye J, Cao L, Yu W. Estimated human papillomavirus vaccine coverage among females 9–45 years of age — China, 2017–2022. China CDC Wkly. 2024;6(19):413–417. doi:10.46234/ccdcw2024.080.
4. World Health Organization. Global strategy to accelerate the elimination of cervical cancer as a public health problem. 2020. https://www.who.int/publications/i/item/9789240014107.
5. Moscicki AB, Schiffman M, Burchell A, Albero G, Giuliano AR, Goodman MT, Kjaer SK, Palefsky J. Updating the natural history of human papillomavirus and anogenital cancers. Vaccine. 2012;5(30 Suppl 05):F24–F33. doi:10.1016/j.vaccine.2012.05.089.
6. World Health Organization (WHO). Expert committee on biological standardization. Guidelines to assure the quality, safety, and efficacy of recombinant HPV virus-like particle vaccines. https://www.who.int/publications/m/item/guidelines-to-assure-the-quality-safety-and-efficacy-of-recombinant-human-papillomavirus-virus-like-particle-vaccines.
7. Basu P, Malvi SG, Joshi S, Bhatla N, Muwonge R, Lucas E, Verma Y, Esmy PO, Poli URR, Shah A, et al. Vaccine efficacy against persistent human papillomavirus (HPV) 16/18 infection at 10 years after one, two, and three doses of quadrivalent HPV vaccine in girls in India: a multicentre, prospective, cohort study. Lancet Oncol. 2021;22(11):1518–1529. doi:10.1016/S1470-2045(21)00453-8.
8. Ferris DG, Samakoses R, Block SL, Lazcano-Ponce E, Restrepo JA, Mehlsen J, Chatterjee A, Iversen O-E, Joshi A, Chu J-L, et al. 4-valent human papillomavirus (4vHPV) vaccine in preadolescents and adolescents after 10 years. Pediatrics. 2017;140(6). doi:10.1542/peds.2016-3947.
9. Goldstone SE, Giuliano AR, Palefsky JM, Lazcano-Ponce E, Penny ME, Cabello RE, Moreira ED, Baraldi E, Jessen H, Ferenczy A, et al. Efficacy, immunogenicity, and safety of a quadrivalent HPV vaccine in men: results of an open-label, long-term extension of a randomised, placebo-controlled, phase 3 trial. Lancet Infect Dis. 2022;22(3):413–425. doi:10.1016/S1473-3099(21)00327-3.
10. Kjaer SK, Nygård M, Sundström K, Dillner J, Tryggvadottir L, Munk C, Berger S, Enerly E, Hortlund M, Ágústsson ÁI, et al. Final analysis of a 14-year long-term follow-up study of the effectiveness and immunogenicity of the quadrivalent human papillomavirus vaccine in women from four Nordic countries. EClinicalMedicine. 2020;23:100401. doi:10.1016/j.eclinm.2020.100401.
11. Kjaer SK, Nygård M, Sundström K, Munk C, Berger S, Dzabic M, Fridrich KE, Waldstrøm M, Sørbye SW, Bautista O, et al. Long-term effectiveness of the nine-valent human papillomavirus vaccine in Scandinavian women: interim analysis after 8 years of follow-up. Hum Vaccines & Immunotherapeutics. 2021;17(4):943–949. doi:10.1080/21645515.2020.1839292.
12. Lehtinen M, Lagheden C, Luostarinen T, Eriksson T, Apter D, Harjula K, Kuortti M, Natunen K, Palmroth J, Petäjä T, et al. Ten-year follow-up of human papillomavirus vaccine efficacy against the most stringent cervical neoplasia end-point—registry-based follow-up of three cohorts from randomized trials. BMJ Open. 2017;7(8):e015867. doi:10.1136/bmjopen-2017-015867.
13. Maldonado I, Plata M, Gonzalez M, Correa, A, Nossa, C., Giuliano, AR, Joura, EA, Ferenczy, A, Ronnett, BM, Stoler, MH, et al. Effectiveness, immunogenicity, and safety of the quadrivalent HPV vaccine in women and men aged 27-45 years. Hum Vaccines Immunotherapeutics. 2022;18(5):2078626. doi:10.1080/21645515.2022.2078626.
14. Mariz FC, Gray P, Bender N, Eriksson T, Kann H, Apter D, Paavonen J, Pajunen E, Prager KM, Sehr P, et al. Sustainability of neutralising antibodies induced by bivalent or quadrivalent HPV vaccines and correlation with efficacy: a combined follow-up analysis of data from two randomised, double-blind, multicentre, phase 3 trials. Lancet Infect Dis. 2021;21(10):1458–1468. doi:10.1016/S1473-3099(20)30873-2.
15. Porras C, Tsang SH, Herrero R, Guillén D, Darragh TM, Stoler MH, Hildesheim A, Wagner S, Boland J, Lowy DR, et al. Efficacy of the bivalent HPV vaccine against HPV 16/18-associated precancer: long-term follow-up results from the Costa Rica vaccine trial. Lancet Oncol. 2020;21(12):1643–1652. doi:10.1016/S1470-2045(20)30524-6.
16. Restrepo J, Herrera T, Samakoses R, Reina JC, Pitisuttithum P, Ulied A, Bekker L-G, Moreira ED, Olsson S-E, Block SL, et al. Ten-year follow-up of 9-valent human papillomavirus vaccine: immunogenicity, effectiveness, and safety. Pediatrics. 2023;152(4):e2022060993. doi:10.1542/peds.2022-060993.
17. Bosch FX, Tsu V, Vorsters A, Van Damme P, Kane MA. Reframing cervical cancer prevention. Expanding the field towards prevention of human papillomavirus infections and related diseases. Vaccine. 2012;30(Suppl 5):1–11. doi:10.1016/j.vaccine.2012.05.090.
18. Saslow D, Andrews KS, Manassaram-Baptiste D, Smith RA, Fontham ETH. Human papillomavirus vaccination 2020 guideline update: American Cancer Society guideline adaptation. CA Cancer J Clin. 2020;70(4):274–280. doi:10.3322/caac.21616.
19. Wang F, Jozkowski KN, Zhang S. Evaluating risk-stratified HPV catch-up vaccination strategies: should we go beyond age 26? Med Decis Mak. 2022;42(4):524–537. doi:10.1177/0272989X211042894.
20. Liu J, Wu L, Bai Q, Ren J. Surveillance for coverage of human papillomavirus (HPV) vaccine and adverse events following immunization with HPV vaccine in Shanghai, 2017-2019. Chin J Vaccines Immun. 2020;26(3):322–325.
21. Zhao FH, Lewkowitz AK, Hu SY, Chen F, Li L-Y, Zhang Q-M, Wu R-F, Li C-Q, Wei L-H, Xu A-D, et al. Prevalence of human papillomavirus and cervical intraepithelial neoplasia in China: a pooled analysis of 17 population-based studies. Int J Cancer. 2012;131(12):2929–2938. doi:10.1002/ijc.27571.
22. Chen W, Zhao Y, Xie X, Liu J, Li J, Zhao C, Wang S, Liao X, Shou Q, Zheng M, et al. Safety of a quadrivalent human papillomavirus vaccine in a phase 3, randomized, double-blind, placebo-controlled clinical trial among Chinese women during 90 months of follow-up. Vaccine. 2019;37(6):889–897. doi:10.1016/j.vaccine.2018.12.030.
23. Wei L, Xie X, Liu J, Zhao Y, Chen W, Zhao C, Wang S, Liao X, Shou Q, Qiu Y, et al. Efficacy of quadrivalent human papillomavirus vaccine against persistent infection and genital disease in Chinese women: a randomized, placebo-controlled trial with

78-month follow-up. Vaccine. 2019;37(27):3617–3624. doi:10.1016/j.vaccine.2018.08.009.
24. Nolan KM, Seaton B, Antonello J, Zhang Y, Cook L, Delfino K, Rubinstein LJ, Cheon K, Group T, Luxembourg A, et al. Development and validation of two optimized multiplexed serologic assays for the 9-valent human papillomavirus vaccine types. mSphere. 2023;8(2):e0096221. doi:10.1128/msphere.00962-21.
25. Opalka D, Matys K, Bojczuk P, Green, T, Gesser, R, Saah, A, Haupt, R, Dutko, F, Esser, MT. Multiplexed serologic assay for nine anogenital human papillomavirus types. Clin Vaccine Immunol. 2010;17(5):818–827. doi:10.1128/CVI.00348-09.
26. Chan ISF, Bohidar NR. Exact power and sample size for vaccine efficacy studies. Commun Stat-Theory Methods. 1998;27(6):1305–1322. doi:10.1080/03610929808832160.
27. Zhao C, Zhao Y, Li J, Li M, Su Y, Mi X, La Tu SY, Shen D, Ren L, Li Y, et al. The eight-year long-term follow-up on the effectiveness of the quadrivalent human papillomavirus vaccine in Chinese women 20-45 years of age. Hum VaccinesImmunotherapeutics. 2022;18(5):2052700. doi:10.1080/21645515.2022.2052700.
28. Donken R, Dobson SRM, Marty KD, Cook D, Sauvageau C, Gilca V, Dionne M, McNeil S, Krajden M, Money D, et al. Immunogenicity of 2 and 3 doses of the quadrivalent human papillomavirus vaccine up to 120 months postvaccination: follow-up of a randomized clinical trial. Clin Infect Dis. 2020;71(4):1022–1029. doi:10.1093/cid/ciz887.
29. Bruni L, Diaz M, Castellsagué X, Ferrer E, Bosch FX, de Sanjosé S, de Sanjosé S. Cervical human papillomavirus prevalence in 5 continents: meta-analysis of 1 million women with normal cytological findings. J Infect Dis. 2010;202(12):1789–1799. doi:10.1086/657321.
30. Mac Eochagain C, Power R, Parker I, Brennan D. HPV vaccination among seropositive, DNA negative cohorts: a systematic review and meta-analysis. J Gynecol Oncol. 2022;33(3):e24. doi:10.3802/jgo.2022.33.e24.
31. Bogani G, Raspagliesi F, Sopracordevole F, Ciavattini A, Ghelardi A, Simoncini T, Petrillo M, Plotti F, Lopez S, Casarin J, et al. Assessing the long-term role of vaccination against HPV after loop electrosurgical excision procedure (LEEP): a propensity-score matched comparison. Vaccines (Basel). 2020;8(4):717. doi:10.3390/vaccines8040717.
32. Petráš M, Dvořák V, Lomozová D, Máčalík, R, Neradová, S, Dlouhý, P, Malinová, J, Rosina, J, Lesná, IK. Timing of HPV vaccination as adjuvant treatment of CIN2+ recurrence in women undergoing surgical excision: a meta-analysis and meta-regression. Sex Transm Infect. 2023;99(8):561–570. doi:10.1136/sextrans-2023-055793.
33. Golia D'augè T, Cuccu I, Etrusco A, D'Amato A, Laganà AS, D'Oria O, Bogani G, Di Donato V, Muzii L, Giannini A, et al. State of the art on HPV-related cervical lesions. Ital J Gynaecol Obstet. 2024;36(2):135. doi:10.36129/jog.2024.161.
34. Stanley M, Pinto LA, Trimble C. Human papillomavirus vaccines–immune responses. Vaccine. 2012;30(Suppl 5):F83–F87. doi:10.1016/j.vaccine.2012.04.106.
35. Prabhu PR, Carter JJ, Galloway DA. B cell responses upon human papillomavirus (HPV) infection and vaccination. Vaccines (Basel). 2022;10(6):837. doi:10.3390/vaccines10060837.
36. Bosch FX, Robles C, Díaz M, Arbyn M, Baussano I, Clavel C, Ronco G, Dillner J, Lehtinen M, Petry K-U, et al. HPV-FASTER: broadening the scope for prevention of hpv-related cancer. Nat Rev Clin Oncol. 2016;13(2):119–132. doi:10.1038/nrclinonc.2015.146.